HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 11 Name: _____ Marks: / 112

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The coordinates of the points P and Q are $(-6,5)$ and $(2,1)$ respectively. P is translated downwards by r units to R.

   (a) Write down, in terms of r, the coordinates of R.

   (b) Find r such that $\angle PQR = 90°$.

(3 marks) (DSE-2026 PI Q03)

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2. The sum of the number of blue pens and the number of red pens in a box is 795. If 123 blue pens are taken out from the box, then the number of blue pens in the box is 25% less than the number of red pens in the box. Find the number of red pens in the box. (4 marks) (DSE-2026 PI Q05)

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3. In Figure 1, ABC is a triangle. D is a point lying on AB. Let E and F be points lying on CB produced and CD produced respectively such that $BD//EF$ and $BC = EF$. It is given that $BD = 8$ cm and $BE = 6$ cm.

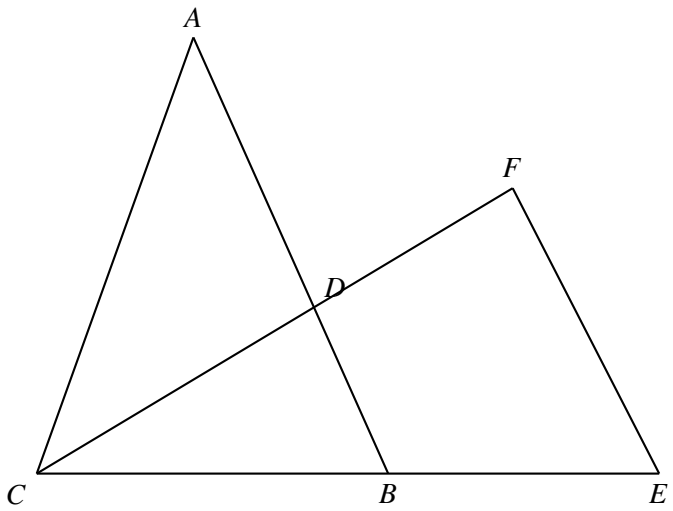


Figure 1

(a) By considering a pair of similar triangles, find the length of BC.

(b) Suppose that $AD = 10$ cm. Prove that $\triangle ABC \cong \triangle CEF$.

(5 marks) (DSE-2026 PI Q08)

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4. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L.

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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5. A solid metal sphere of radius 10 cm is melted and recast into three similar solid right circular cylinders X, Y and Z. Let r_1 cm, r_2 cm and r_3 cm be the base radii of X, Y and Z respectively. It is given that $r_1 : r_2 : r_3 = 2 : 3 : 5$.

(a) Express the volume of Y in terms of π.

(b) A craftsman finds that the height of Y is 9 cm.

(i) Find r_2.

(ii) The craftsman claims that the total surface area of Z is greater than the surface area of the sphere. Do you agree? Explain your answer.

(8 marks) (DSE-2026 PI Q13)

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6. When the polynomial $g(x)$ is divided by x^2+2x+a, the quotient and the remainder are $x+13$ and $b(x-3)$ respectively, where a and b are constants. It is given that $g(x)$ is divisible by $x-3$.

(a) Find a.

(b) Is there only one possible value of b such that at least two roots of the equation $g(x)=0$ are equal? Explain your answer.

(8 marks) (DSE-2026 PI Q14)

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7. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

(a) the probability that exactly 2 red balls are drawn;

(b) the probability that exactly 2 red balls or exactly 2 white balls are drawn.

(4 marks) (DSE-2026 PI Q15)

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8. It is given that P, Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C.

(b) The tangent to C at P and the tangent to C at R intersect at the point H. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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9. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

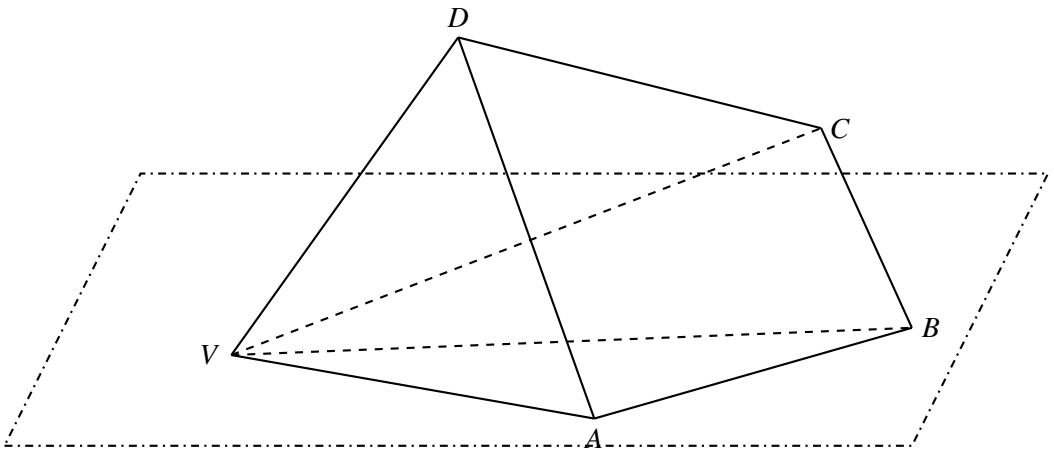


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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10. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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11. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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12. Solve $(x+2s)(x+t) = sx+st$, where s and t are constants.

A. $x = -s$

B. $x = -2s$

C. $x = -s$ or $x = -t$

D. $x = -2s$ or $x = -t$

(DSE-2026 PII Q06)

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13. Let $f(x) = (x+d)(x-2)+9x$, where d is a constant. If $f(1+d) = 0$, then $d =$

A. -2.

B. -1.

C. 1.

D. 2.

(DSE-2026 PII Q07)

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14. Let $g(x) = hx^3 + 2x^2 + kx - 5$, where h and k are constants. If $g(x)$ is divisible by $x - 4$, find the remainder when $g(x)$ is divided by $x + 4$.

A. -27

B. -10

C. 0

D. 54

(DSE-2026 PII Q08)

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15. The costs of rice of brand S and rice of brand T are \$12/kg and \$18/kg respectively. If s kg of rice of brand S and t kg of rice of brand T are mixed so that the cost of the mixture is \$16/kg, then $s : t =$

A. 1 : 2.

B. 2 : 1.

C. 2 : 3.

D. 3 : 2.

(DSE-2026 PII Q10)

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16. The perimeter and the area of a sector are k cm and $3k$ cm^2 respectively. If the radius of the sector is 14 cm, then the arc length of the sector is

A. 19 cm.

B. 21 cm.

C. 28 cm.

D. 49 cm.

(DSE-2026 PII Q16)

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17. It is given that $\triangle PQR$ is a right-angled triangle with $\angle QPR = 90^\circ$. Denote the area of $\triangle PQR$ by S. If $QR = p$, $PR = q$ and $\angle PQR = \theta$, which of the following must be true?

A. $S = \frac{1}{2}pq$

B. $S = \frac{1}{2}pq\sin\theta$

C. $S = \frac{1}{2}pq\cos\theta$

D. $S = \frac{1}{2}pq\tan\theta$

(DSE-2026 PII Q19)

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18. It is given that $ABCDEF$ is a regular 6-sided polygon. If AC and BF intersect at the point G, which of the following are true?

I. $BG//CE$

II. $\triangle ABG \sim \triangle BDC$

III. $\triangle AGF \cong \triangle BGC$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

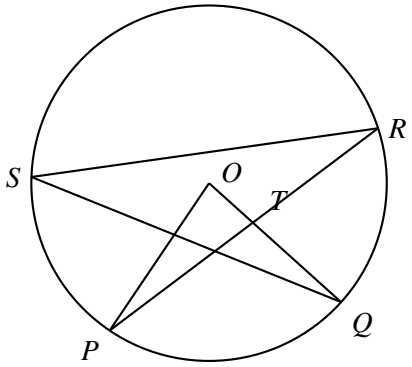
(DSE-2026 PII Q23)

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19. In the figure, O is the centre of the circle $PQRS$. PR and OQ intersect at the point T. If $\angle POT = 82^\circ$ and $\angle PTQ = 104^\circ$, then $\angle QSR =$



A. 22°.

B. 27°.

C. 30°.

D. 49°.

(DSE-2026 PII Q25)

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20. The equations of the straight lines L_1 and L_2 are $3x + 5y = 6$ and $6x - 10y = 13$ respectively. Let P be a moving point in the rectangular coordinate plane such that the perpendicular distance from P to L_1 is equal to the perpendicular distance from P to L_2. The locus of P is

A. a point.

B. a circle.

C. a square.

D. a pair of straight lines.

(DSE-2026 PII Q26)

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21. The x-coordinate of the centre of the circle C is 5. If C passes through the points $(5,0)$ and $(11,4)$, then the circumference of C is

A. 13π.

B. 14π.

C. 22π.

D. 42π.

(DSE-2026 PII Q27)

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22. Consider the following integers:

$$4 \quad 7 \quad 7 \quad 7 \quad 8 \quad 11 \quad 11 \quad 13 \quad x \quad y$$

Let u, v and w be the variance, the median and the mode of the above integers respectively. If both the mean and the range of the above integers are 9, which of the following must be true?

I. $u < 7$

II. $v > 7$

III. $w = 7$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2026 PII Q30)

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23. $1010100000001110_2 =$

A. $21 \times 2^{11} + 14$.

B. $21 \times 2^{11} + 28$.

C. $21 \times 2^{12} + 14$.

D. $21 \times 2^{12} + 28$.

(DSE-2026 PII Q31)

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24. It is given that $\log y$ is a linear function of x. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 2 and 1 respectively. If $y = ab^x$, find a.

A. 0.01

B. 0.1

C. 10

D. 100

(DSE-2026 PII Q36)

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25. Consider the following system of inequalities:

$$\begin{cases} y \leq 7 \\ 3x - 4y \leq 2 \\ 6x + 13y \geq 25 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in R, then the least value of $5x + 4y + 28$ is

A. 1.

B. 19.

C. 42.

D. 106.

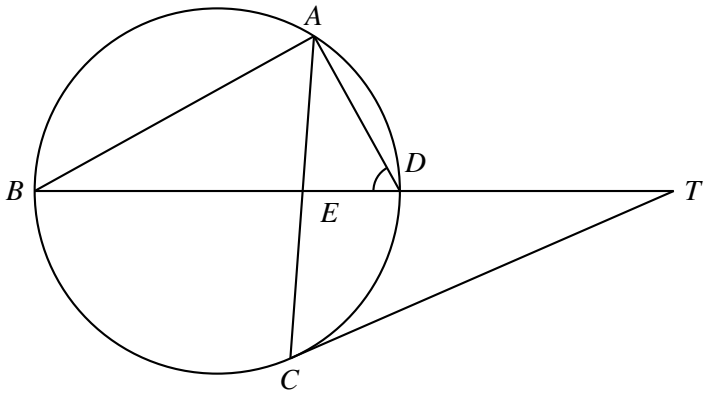
(DSE-2026 PII Q37)

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26. In the figure, $ABCD$ is a circle. AC and BD intersect at the point E. BD is produced to the point T such that TC is tangent to the circle. It is given that $AB//TC$. If $\angle AED = 85°$ and $\angle CTE = 23°$, then $\angle ADE =$



A. 39°.

B. 56°.

C. 62°.

D. 72°.

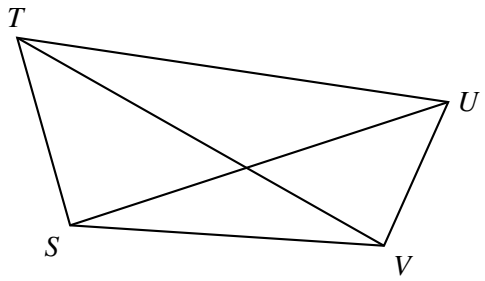
(DSE-2026 PII Q38)

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27. The figure shows the quadrilateral $STUV$, where $ST = 8$ cm and $SV = 5$ cm. If $\angle UTV = 30°$ and $\angle TSU = \angle TVU = 90°$, find $\angle SUV$ correct to the nearest degree.



A. 22°

B. 28°

C. 30°

D. 38°

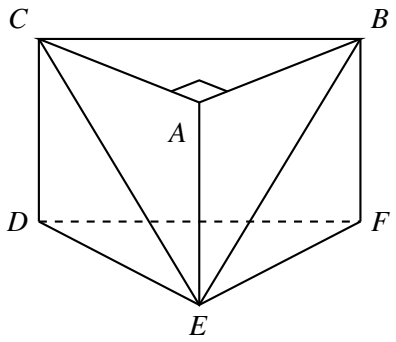
(DSE-2026 PII Q39)

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28. The figure shows the right prism $ABCDEF$, where $AB : AC : AE = 1 : 1 : 2$. Let θ be the angle between the plane ABC and the plane BCE. If $\angle BAC = 90°$, find $\sin\theta$.



A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{\sqrt{2}}{4}$

D. $\frac{2\sqrt{2}}{3}$

(DSE-2026 PII Q40)

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29. For $90^\circ \leq \theta \leq 360^\circ$, how many roots does the equation $6\sin^4\theta - 5\sin^2\theta + 1 = 0$ have?

A. 2

B. 4

C. 6

D. 8

(DSE-2026 PII Q41)

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30. There are 5 full-time employees and 3 part-time employees in a company. From the company, 7 employees are selected to form a queue. If no part-time employees are next to each other, how many different queues can be formed?

A. 5040

B. 14400

C. 18000

D. 32400

(DSE-2026 PII Q43)

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31. The table below shows the scores (in marks) and the corresponding standard scores of four students in a test.

Score (marks)	α	40	58	β
Standard score	-1	0	2	3

Find $\beta - \alpha$.

A. 31

B. 36

C. 54

D. 98

(DSE-2026 PII Q44)

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32. If the standard deviation of the five numbers p, q, r, s and t is 5, then the variance of the five numbers $\frac{p+2}{5}$, $\frac{q+2}{5}$, $\frac{r+2}{5}$, $\frac{s+2}{5}$ and $\frac{t+2}{5}$ is

A. 1.

B. 5.

C. 9.

D. 25.

(DSE-2026 PII Q45)

I can finish _____ questions

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